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About Me

I enjoy building useful products that meet real customer needs. I've spent over a decade designing and developing software. At Haptiq, I lead a 5-engineer AI team delivering LLM-native products for private equity and credit research, from requirements and architecture through production delivery.

Experience *(Selected)*

AI Engineer III (Technical Lead)**,** *Haptiq LLC* 2025 – Present

- Owned LLM development across Haptiq's enterprise operations (Orion) and asset management (Olympus) platforms, leading a **5-engineer AI team** directly under the CTO.
- Proposed, architected, and led development of the **agentic AI assistant** for Orion, improving text to SQL performance by over *30%* while attaining faster TTFT and more flexible orchestration.
- Led development of **AI-powered research and analysis** for Olympus, including a RAG retrieval pipeline spanning *30K+* financial research documents.
- Helped drive **millions in contract revenue** through enterprise adoption of Olympus by investment firms managing **billions in assets**.
- Established team standards for Git workflows, code review, and CI/CD.

Research Assistant, *University of Waterloo* 2019 – 2025

Scalable Type Inference on Java Code Snippets + LLM Evaluation

- Designed *novel evaluation* framework for LLM-based type inference on code snippets, identifying a critical *data leakage* issue in prior evaluations.
- Developed a novel type inference method leveraging semantic information in code structures, improving runtime performance by *over 20x*, while maintaining high accuracy
- Optimized SQL database queries with tables containing *millions* of Java types and methods.

Bash Empirical Study

- Published a large-scale empirical study analyzing 1M+ Bash scripts from GitHub.
- Manually analyzed bug reports to classify common bug patterns.

Teaching Assistant, *University of Waterloo* 2019 – 2025

- Courses: App Dev (CS346), Software Design (CS446), Software Testing (SE465), Sequential Programs (CS241), Object Oriented Programming (CS246), Functional Programming (CS135)

Technical Knowledge *(Selected)*

Languages	Python, Java, JavaScript, SQL, Bash
Technologies	Pydantic AI, FastAPI, Langfuse, LangGraph, RAG, MCP, MongoDB, ANTLR, Spring
Other	Git, GitHub Actions, Docker, Azure App Service, Claude, Codex, Cursor

Education

University of Waterloo, Waterloo, ON 2019 – 2025
Ph.D. in Computer Science – Advisor: Chengnian Sun (GPA: 90.75/100)

Northeastern University, Boston, MA 2016 – 2018
B.S. in Computer Science – Minor in Computer Engineering (GPA: 3.9/4.0)

Publications *(Selected)*

Scitix: Scalable Constraint-Based Type Inference for Code Snippets with Unknown Types. ISSTA 26
Evaluating the Genuine Type Inference Capabilities of LLMs for Java Code Snippets. TOSEM 26
LPR: Large Language Models-Aided Program Reduction. ISSTA 24
Revisiting the Evaluation of Deep Learning-Based Compiler Testing. IJCAI 23
Bash in the Wild: Language Usage, Code Smells, and Bugs. TOSEM 22